Mechanics
Mechanics.SchDoc

Power VCC
Power_VCC.SchDoc
ON_REQ
VIN
OUT_LV
VCC
V_VCC
TP3
VCC
V_VCC

Power 15V
Power_15V.SchDoc
VCC
+15V
V_15V
+15V
TP1
V_15V

Power 3V3
Power_3V3.SchDoc
+3V3
V_3V3
+3V3
TP2
V_3V3

Power 5V
Power_5V.SchDoc
+5V
V_5V
+5V
TP4
V_5V

Power Analog
Power_Analog.SchDoc
+3V3_A
V_3V3_A
+3V3_A
TP10
V_3V3_A

HV_DET

TP5 GND_PWR
TP6 GND
TP7 GND
TP8 GND
TP9 GND

OVP
Overvoltage_Protection.SchDoc
OUT_LV
OVP_RESET
OVP/

OCP
OverCurrent_Protection.SchDoc
I_ISO
OCP/

OUT_LV

Battery & BMS
Battery_BMS.SchDoc
VBAT_STBY
+3V3_STBY
VBAT
BMS_ON_OFF
BMS_CTRL
I2C
VBAT_STBY
+3V3_STBY

On Off Controller
On_Off.SchDoc
VBAT
BMS_ON_OFF
UI_ON_OFF
ON_REQ
VIN
V_IN
VBAT
BMS_ON_OFF
UI_ON_OFF

TP11
VIN

Converter
Converter.SchDoc
VIN
OVP/
CONV_CTRL
DISCHARGE/
OUT_LV

TP12

Output terminals
Output_Terminals.SchDoc
OUT_LV
OUT_HV
HV_DET
SENS_P
SENS_N
TERM_P
TERM_N
I_ISO_P
I_ISO_N
I_OUT_P
I_OUT_N

Converter High Voltage
Converter_HV.SchDoc
VIN
OCP/
HV_CTRL
OUT_HV
V_HV

OUT_HV
TP13

Measurement
Measurement.SchDoc
SENS_P
SENS_N
TERM_P
TERM_N
I_ISO_P
I_ISO_N
I_OUT_P
I_OUT_N
ADC
SPI
I_ISO
V_TERM
I_OUT

V_IN
ON_REQ
OCP/
OVP/
V_HV

User Interface
User_Interface.SchDoc
BMS_ON_OFF
HV_DET
UI_ON_OFF
I2C
DISPLAY
BUTTON
ENCODER
UART
UI_PRESENT

Microcontroller
Microcontroller.SchDoc
V_HV
OVP/
OCP/
V_IN
BMS_CTRL
I2C
DISPLAY
BUTTON
ENCODER
UART
UI_PRESENT
ON_REQ
OVP_RESET
CONV_CTRL
DISCHARGE/
HV_CTRL
OUT_SEL_HV
OUT_SEL_ISO
I_1A_REF
SHUNT_EN
SHUNT_ISO_EN
V_VCC
V_15V
V_5V
V_3V3
V_3V3_A
QSPI
SWD
VREF_UC
VREF/2_UC
RESET/
BOOT0
I_ISO
V_TERM
I_OUT
V_OUT
TEMP_CURRENT
ADC
SPI

BMS_CTRL
I2C
DISPLAY
BUTTON
ENCODER
UART
UI_PRESENT

OUT_SEL_HV
OUT_SEL_ISO
I_1A_REF
SHUNT_EN
SHUNT_ISO_EN
V_OUT
I_OUT
TEMP_CURRENT

V_VCC
V_15V
V_5V
V_3V3
V_3V3_A

Microcontroler Peripherals
Peripherals.SchDoc
I2C
UART
QSPI
SWD
VREF_UC
VREF/2_UC
RESET/
BOOT0
VREF
VREF/2

TEMP_CURRENT
V_OUT
I_OUT
V_TERM
I_ISO

Assembly variants:

PS1-1, full funcionallity:
- 60V / 1A
- 5V / 10A
- Battery charging
- Voltmeter (1000V)
- Amperemeter (10A)
- Isolation resistance measurement (1000V)
- Low range resistance measurement (1A)

PS1-10, reduced functionallity:
- 60V / 1A
- 5V / 10A
- Battery charging

M3 SERIAL 2D
PS1-1B M1 PCB NAME
LOGO M4 LOGO WHE
M6 QR PROJ URL
LOGO M7 LOGO FSG
M2 Tooling JLC
M5 Tooling JLC
FM1 FM2 FM3 FM4 FM5 FM6

| | | |
|---|---|---|
| ***Top*** | | * |
| PS1-1BA | Battery Source | * |
| Date: 2026-Mar-05 | Sheet 1 of 20 | Size: A3 * |
| Schema: ahorat / daniw | Layout: daniw | PS1-1-Top.SchDoc |

5S

4S

TP14 TP15 TP16 TP17 TP18

R3 R5 R7 R8 R9 R10
100R/0603

BAT5 BAT6 BAT7 BAT8 BAT9
BAT_LFP-26700-4.5AH

BAT1 BAT2 BAT3 BAT4
BAT_LFP-26700-4.5AH

C4 C3 C2 C1

C2 C3 C5 C7 C8
100N/0603/50V/X7R

C9
100N/0603/50V/X7R

BAT-

VC5 VC4 VC3 VC2 VC1 VC0

U1
BQ76905
18 VC5
19 VC4A
20 VC4B
1 VC3A
2 VC3B
3 VC2
4 VC1
5 VC0
8 TS
11 VSS EP
17 BAT
16 REGSRC
15 REGOUT
14 ALERT
12 SCL
13 SDA
10 CHG
9 DSG
7 SRN
6 SRP

D1 D2
B5819W SL

R1 R2
10R/0603

C1
1U/0603/50V/X5R

BAT-

C4
1U/0603/50V/X5R

BAT-

VBAT 3

Q1
BSS84

VBAT_STBY

R4
10M/0603

R6
1M/0603

+3V3_STBY

C6
100N/0603/50V/X7R

BAT-

I2C Address:
0b0001000 (0x08)

I2C
ALERT
SCL
SDA

I2C 12, 13, 14

BMS_ON_OFF
CHG
DSG

BMS_ON_OFF 3, 12

Q2
BSS138

R11
100R/0603

C10
100N/0603/50V/X7R

R13
100R/0603

R12
10M/0603

R14
0R005/2512

NT1
NET-TIE_1000X100-I4

GND GND_PWR

Adjustment to change from 4S to 5S configuration:
- remove connection between C2 and C3 (red)
- Change text "C3" to net label
- Change BAT2 to BAT_LFP-26700-4.5AH

TS

J1
Solder Jumper 2pin 0.3mm

R15
10K/40MM/NTC/3380K

C11
1N/0603/50V/X7R

R16
10K/0603/NTC/3380K

BAT-

SMD thermistor for testing purposes only

+3V3_A

U2B
TLV314IDBV
VCC
VEE

C13
100N/0603/50V/X7R

GND

VREF/2

R17
330K/0603

C12
1N/0603/50V/X7R

R18
10K/0603

U2A
TLV314IDBV

R19
10K/0603

R20
330K/0603

C14
1N/0603/50V/X7R

BMS_CTRL
I_BAT
WAKEUP

BMS_CTRL 13

D3
B5819W SL

D4
DDZ9678

R21
1K/0603

BMS_CTRL.WAKEUP

| | | | |
|---|---|---|---|
| ***Battery & BMS*** | | | * |
| PS1-1BA | Battery Source | | * |
| Date: 2026-Mar-05 | Sheet 2 of 20 | Size: A3 | * |
| Schema: ahorat / daniw | Layout: daniw | Battery_BMS.SchDoc | |

Minimum value for R21 :
4.7R due to power limit
with package 0603

Q3
AGM40P35AP

R22
27R/0603

R23
27R/0603

R24
27R/0603

R25
27R/0603

Q4
MMBT4403L

R26
100K/0603

Q5
AGM40P35AP

Q6
AGM40P35AP

2 VBAT

VIN 4, 5, 15

VBAT_STBY

VBAT_STBY

R30
100K/0603

Q9
BSS84

D5
B5819W SL

Q7
BSS84

R27
100K/0603

Q8
BSS84

R28
100K/0603

R31
1M/0603

D6
BZT52C15T

R29
200K/0603

V_IN 13

U3B
TLV1701AIDBV

C16
100N/0603/50V/X7R

R35
1M/0603

R36
10M/0603

R37
100K/0603

R38
10K/0603

R32
10K/0603

R33
100K/0603

Q10
BSS138

Q11
BSS138

R34
10K/0603

C15
10N/0603/50V/X7R

GND

D7
BAV199

U3A
TLV1701AIDBV

ON_OFF

OFF

Q13
BSS138

R39
6M8/0603

R43
100K/0603

R44
100K/0603

C17
1U/0603/50V/X5R

R42
1M/0603

Q12
BSS138

R40
10K/0603

R41
100K/0603

12 UI_ON_OFF

ON

UI_ON_OFF.ON

Q14
BSS138

R45
100K/0603

13, 15 ON_REQ

ON_REQ

Q15
BSS138

R46
100K/0603

BMS_ON_OFF

DSG

Q16
BSS138

2, 12 BMS_ON_OFF

CHG

Q17
BSS138

| | | | |
|---|---|---|---|
| ***ON / OFF Controller*** | | | * |
| PS1-1BA | Battery Source | | * |
| Date: 2026-Mar-05 | Sheet 3 of 20 | Size: A3 | * |
| Schema: ahorat / daniw | Layout: daniw | On_Off.SchDoc | |

3, 5, 15 VIN

OUT_LV 7, 9, 15

C18 22U/0805/25V/X5R — GND_PWR
C19 22U/0805/25V/X5R — GND_PWR
C20 22U/0805/25V/X5R — GND_PWR
C21 22U/0805/25V/X5R — GND_PWR

C22 2U2/1206/100V/X7R — GND_PWR
C23 2U2/1206/100V/X7R — GND_PWR
C24 2U2/1206/100V/X7R — GND_PWR
C25 2U2/1206/100V/X7R — GND_PWR
C26 2U2/1206/100V/X7R — GND_PWR
C27 2U2/1206/100V/X7R — GND_PWR
C28 2U2/1206/100V/X7R — GND_PWR
C29 2U2/1206/100V/X7R — GND_PWR
C30 2U2/1206/100V/X7R — GND_PWR
C31 2U2/1206/100V/X7R — GND_PWR
C32 2U2/1206/100V/X7R — GND_PWR
C33 2U2/1206/100V/X7R — GND_PWR
C34 2U2/1206/100V/X7R — GND_PWR
C35 2U2/1206/100V/X7R — GND_PWR
C36 2U2/1206/100V/X7R — GND_PWR

0: Latch active
1: Normal operation

7, 13 OVP/

D8
B5819W SL

C37
1U/0603/50V/X5R
+5V
GND

U4
LMG2100R044
VCC, VIN, HB, NC, HS, SW, HI, LI, AGND, PGND
+5V
C39
1U/0603/50V/X5R
LV_SW_PRIM
GND
GND_PWR

U5
LMG2100R044
VIN, VCC, HB, NC, HS, SW, HI, LI, PGND, AGND
C40
1U/0603/50V/X5R
LV_SW_SEC
+5V
+5V
C38
1U/0603/50V/X5R
GND
GND_PWR
GND

L1
10UH/MDA1365

L2
10UH/PQ2012J

R48
0R/0603
GND_PWR

R49
0R/0603
GND

CONV_CTRL
PRIM_H
PRIM_L
SEC_L
EN
SEC_H

U6C
SN74LVC08A TSSOP
CONV_PRIM_H

U6D
SN74LVC08A TSSOP
CONV_PRIM_L

U6A
SN74LVC08A TSSOP
CONV_SEC_L

U6B
SN74LVC08A TSSOP
CONV_SEC_H

13 CONV_CTRL

R47
100K/0603

R50
100K/0603
GND

+3V3
C41
100N/0603/50V/X7R
GND

+3V3
U6E
SN74LVC08A TSSOP
VCC
GND
GND

VREF

R51
10K/0603
TEMP_PRIM
TEMP_SEC
TEMP_TRAFO
C42
10N/0603/50V/X7R
R54
10K/0603/NTC/3380K
Inductor temperature

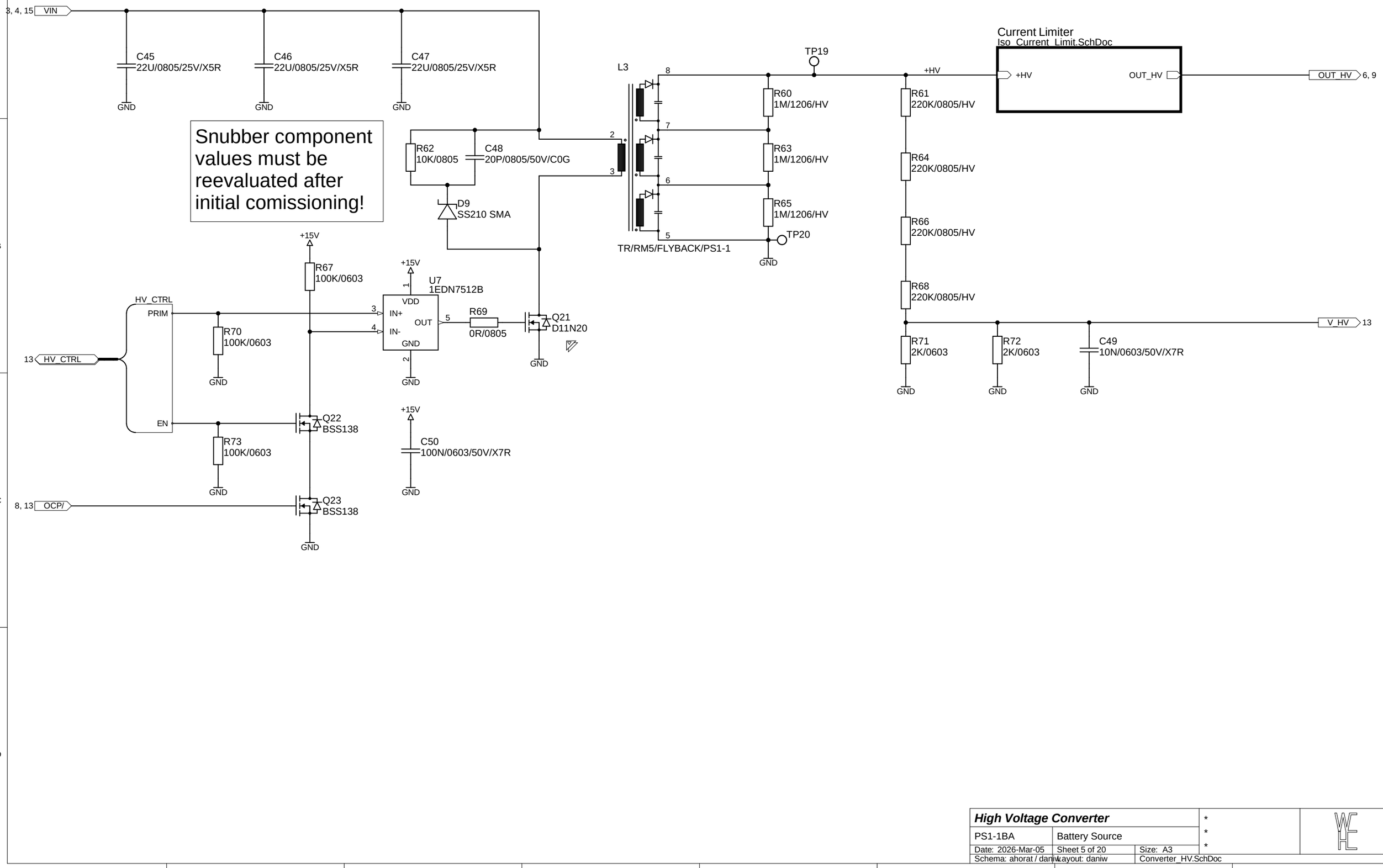
R52
10K/0603
C43
10N/0603/50V/X7R
R55
10K/0603/NTC/3380K
Primary MOSFET temperature

R53
10K/0603
C44
10N/0603/50V/X7R
R56
10K/0603/NTC/3380K
Secondary MOSFET temperature

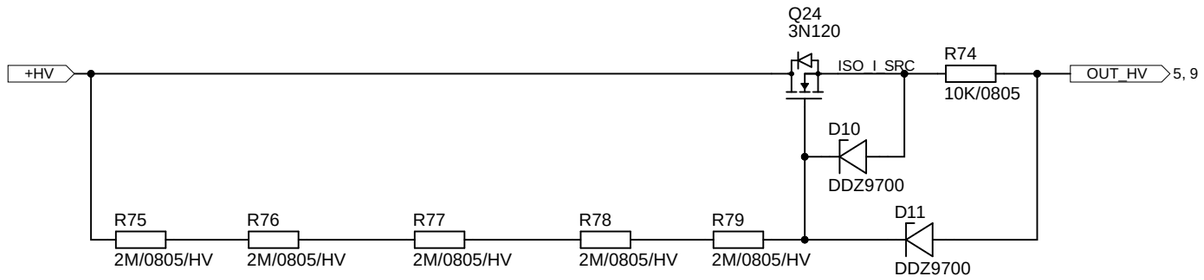
13 DISCHARGE/

+15V
R57
100K/0603
Q18
D11N20
Q19
BSS138
Q20
MMBT3904
R58
100K/0603
R59
30R/0603

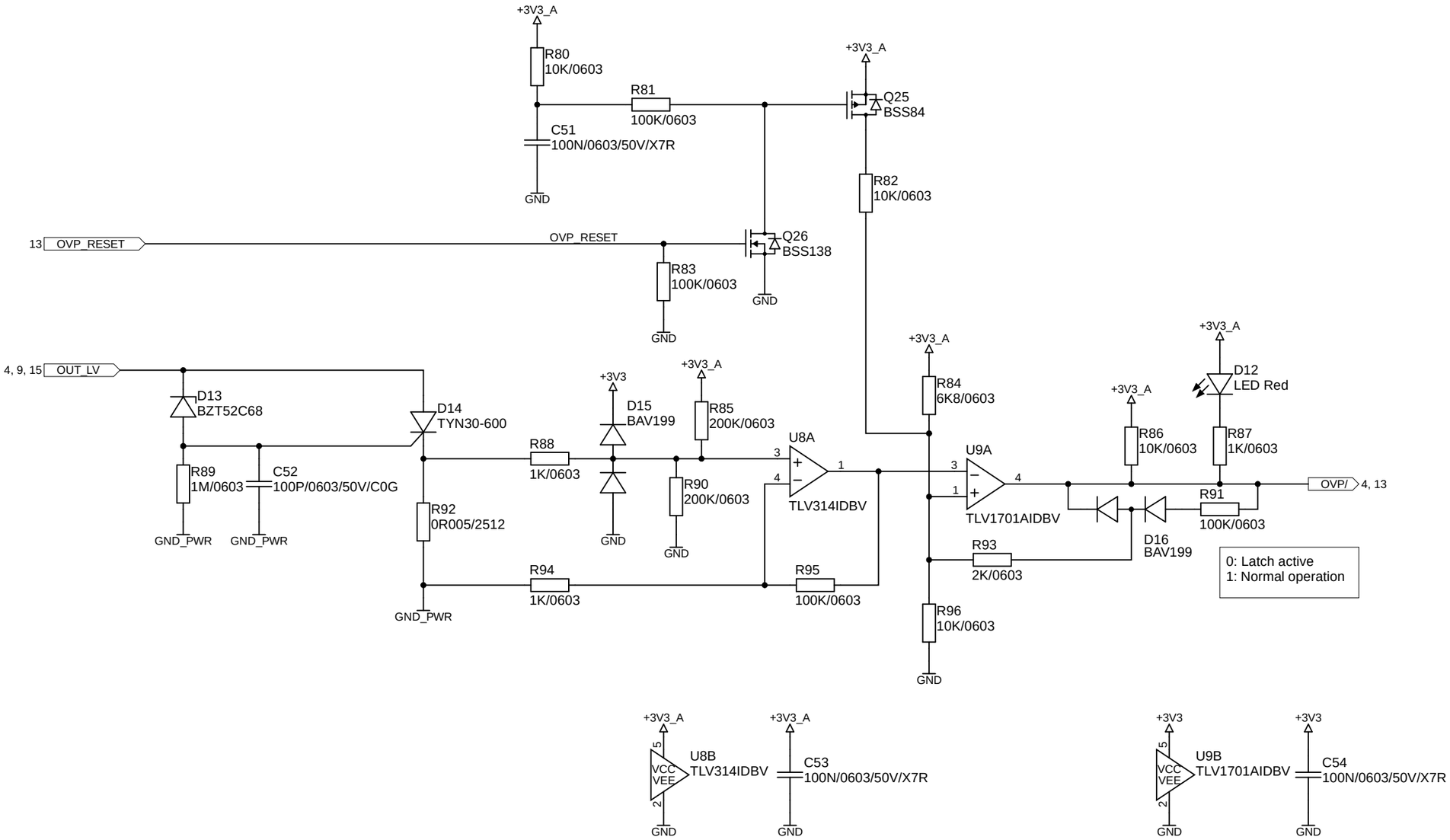
| Converter | | |
|---|---|---|
| PS1-1BA | Battery Source | |
| Date: 2026-Mar-05 | Sheet 4 of 20 | Size: A3 |
| Schema: ahorat / daniw | Layout: daniw | Converter.SchDoc |



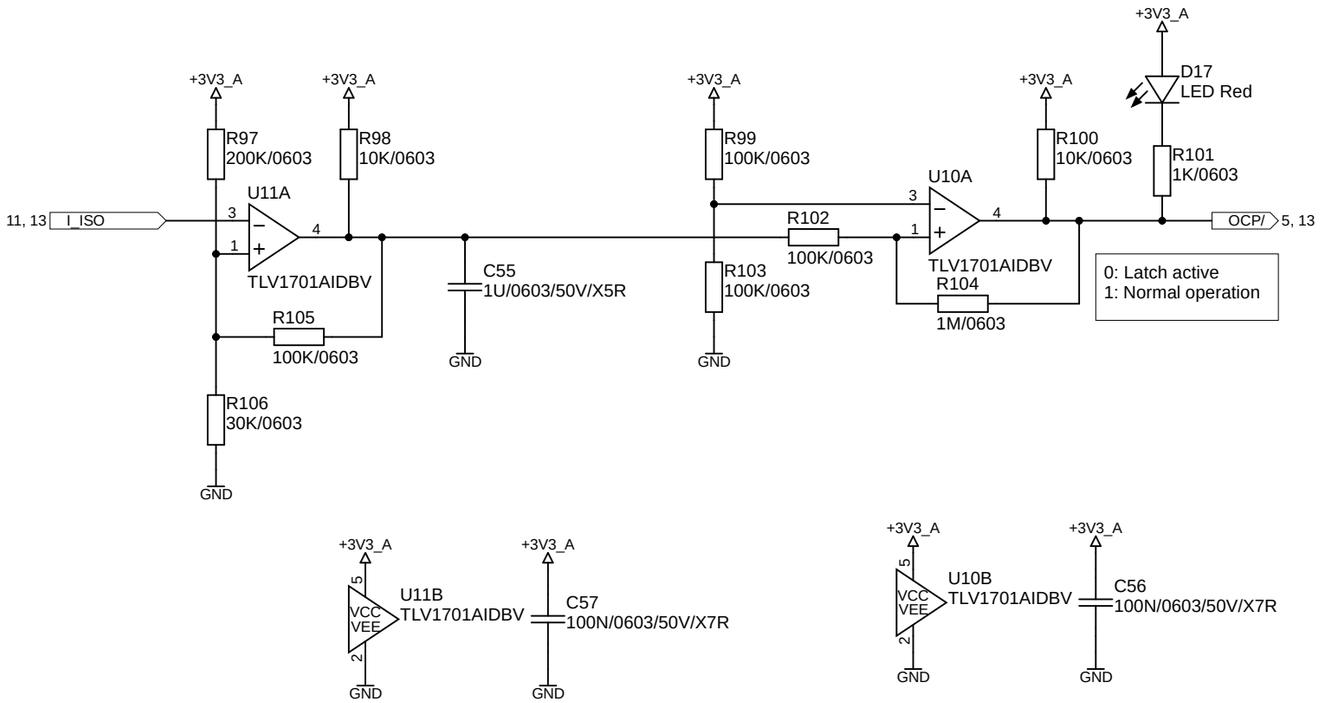

| | | | |
|---|---|---|---|
| ***High Voltage Converter*** | | | * * * |
| PS1-1BA | Battery Source | | |
| Date: 2026-Mar-05 | Sheet 5 of 20 | Size: A3 | |
| Schema: ahorat / daniw | Layout: daniw | Converter_HV.SchDoc | |



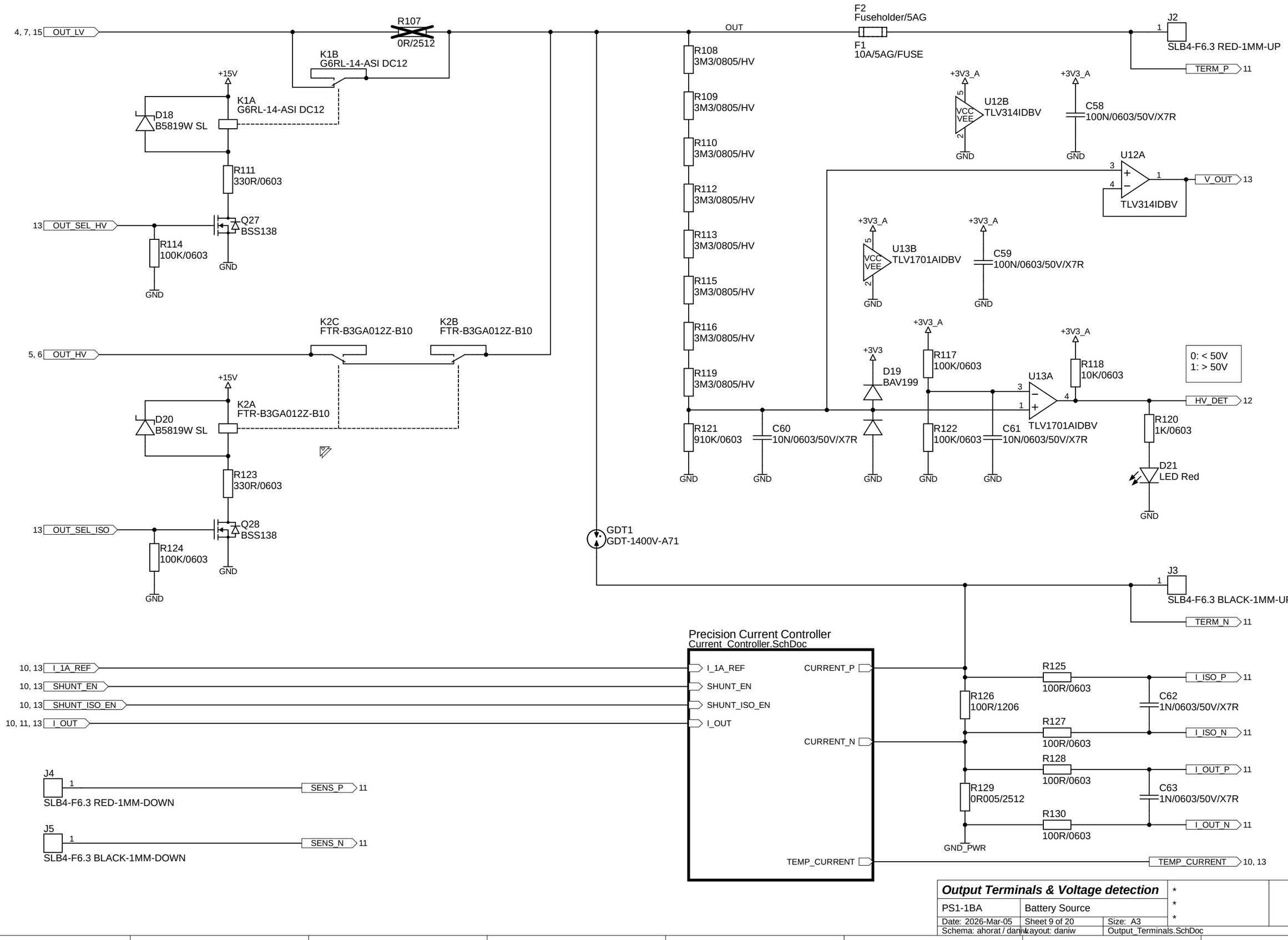

| | | | |
|---|---|---|---|
| ***Current Limiter*** | | | * * * |
| PS1-1BA | Battery Source | | |
| Date: 2026-Mar-05 | Sheet 6 of 20 | Size: A3 | |
| Schema: ahorat / daniw | Layout: daniw | Iso_Current_Limit.SchDoc | |



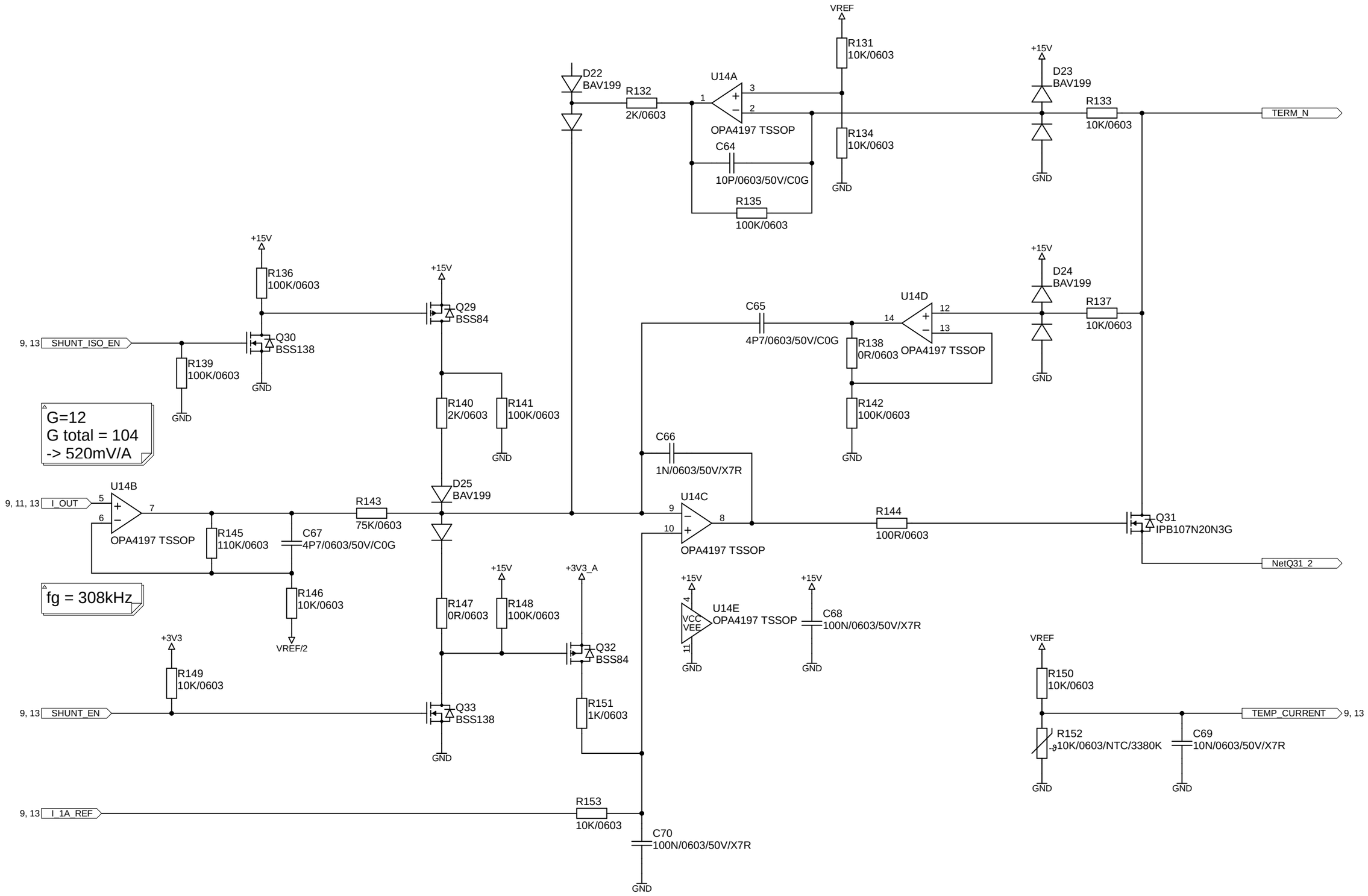

| | | |
|---|---|---|
| ***Overvoltage Protection*** | | * |
| PS1-1BA | Battery Source | * |
| Date: 2026-Mar-05 | Sheet 7 of 20 | Size: A3 * |
| Schema: ahorat / daniw | Layout: daniw | Overvoltage_Protection.SchDoc |



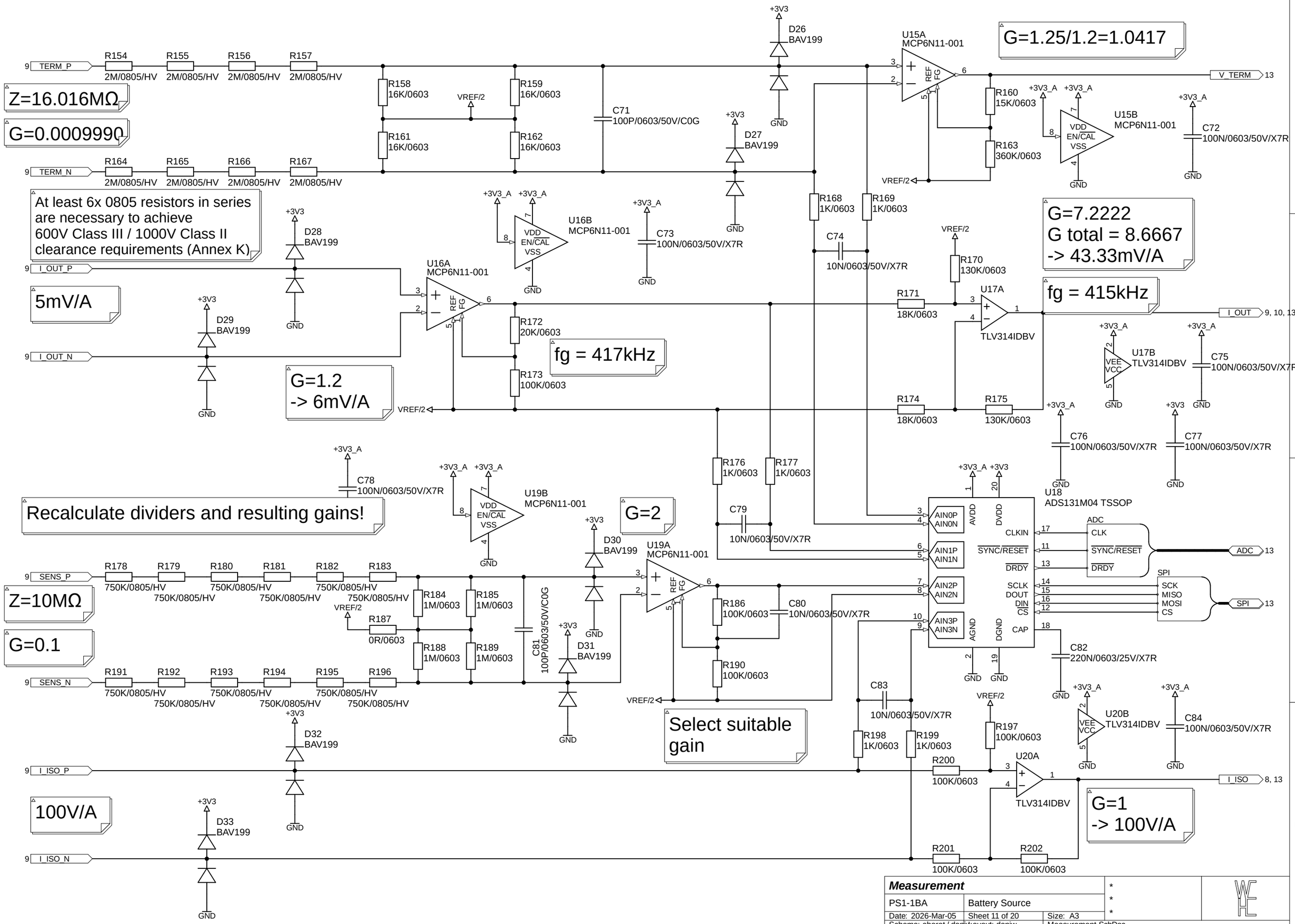

| | | |
|---|---|---|
| ***Protection*** | | * * * |
| PS1-1BA | Battery Source | |
| Date: 2026-Mar-05 | Sheet 8 of 20 | Size: A3 |
| Schema: ahorat / daniw | Layout: daniw | Overcurrent_Protection.SchDoc |

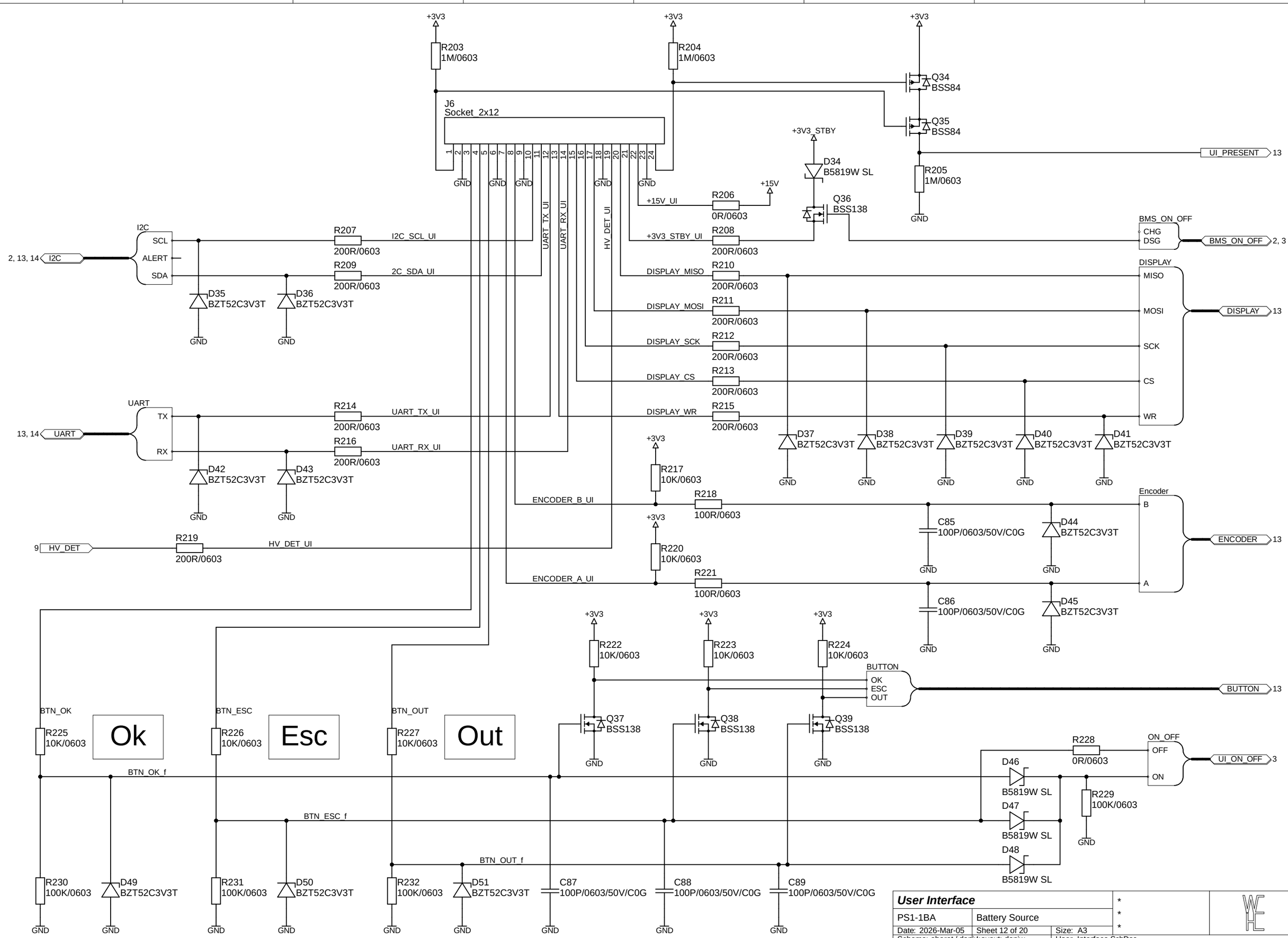


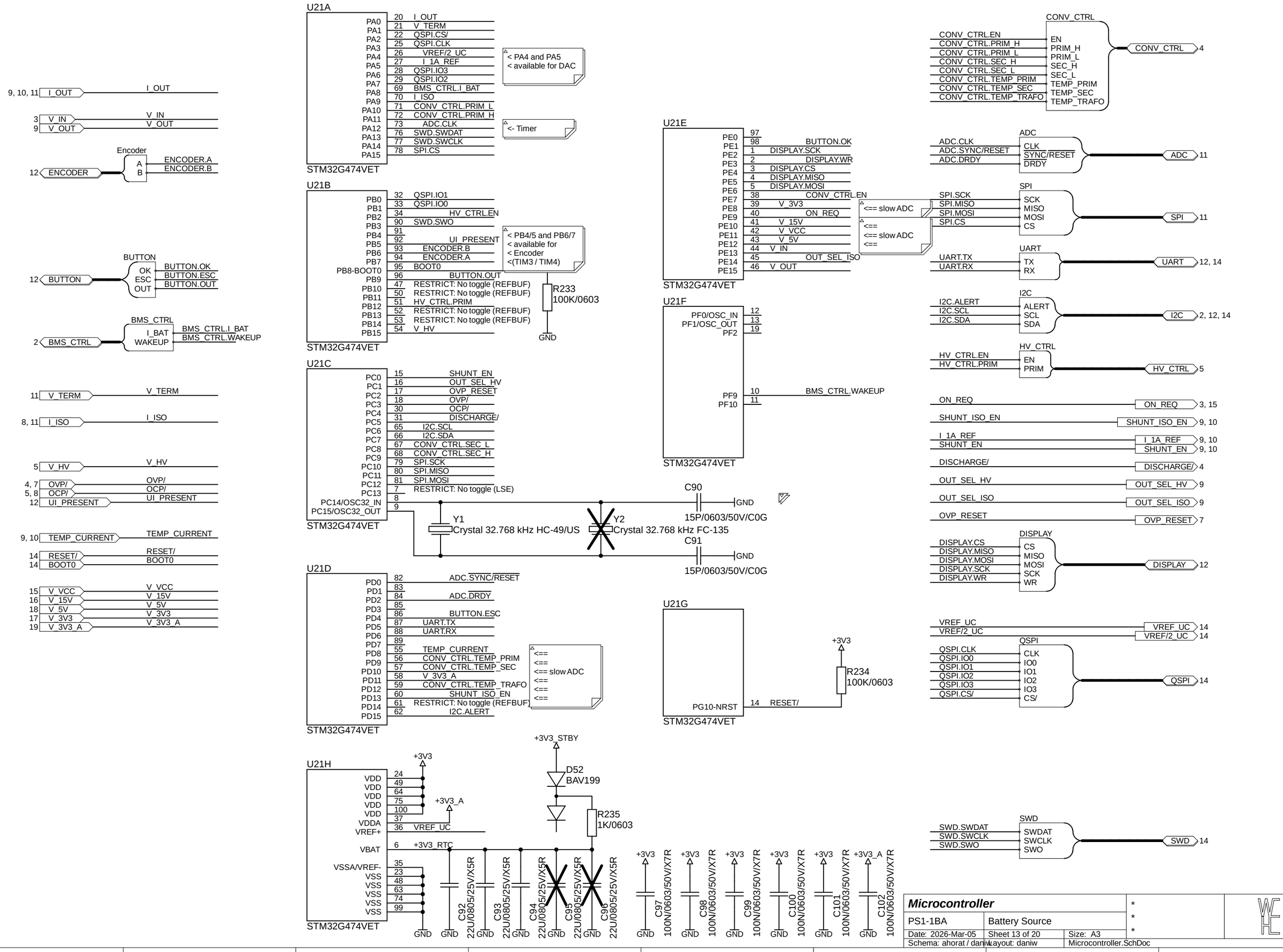

| | | | |
|---|---|---|---|
| ***Output Terminals & Voltage detection*** | | | * * * |
| PS1-1BA | Battery Source | | |
| Date: 2026-Mar-05 | Sheet 9 of 20 | Size: A3 | |
| Schema: ahorat / daniw | Layout: daniw | Output_Terminals.SchDoc | |

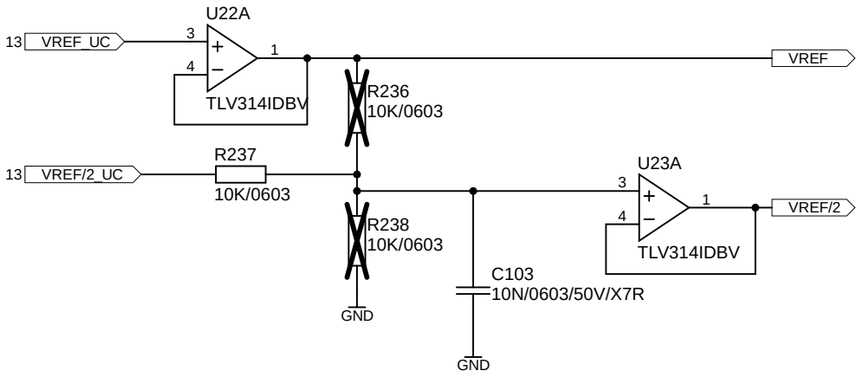


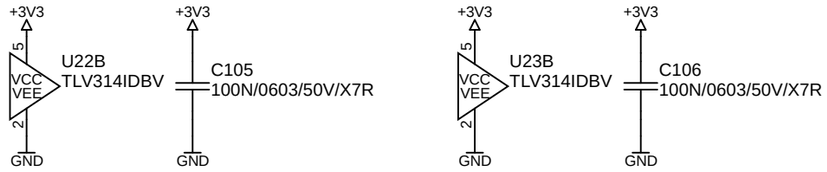

| | | | |
|---|---|---|---|
| ***Precision Current source*** | | | * |
| PS1-1BA | Battery Source | | * |
| Date: 2026-Mar-05 | Sheet 10 of 20 | Size: A3 | * |
| Schema: ahorat / daniw | Layout: daniw | Current_Controller.SchDoc | |


TERM_P
TERM_N
R154 2M/0805/HV
R155 2M/0805/HV
R156 2M/0805/HV
R157 2M/0805/HV
R164 2M/0805/HV
R165 2M/0805/HV
R166 2M/0805/HV
R167 2M/0805/HV
Z=16.016MΩ
G=0.0009990
At least 6x 0805 resistors in series are necessary to achieve 600V Class III / 1000V Class II clearance requirements (Annex K)
R158 16K/0603
R159 16K/0603
R161 16K/0603
R162 16K/0603
VREF/2
C71 100P/0603/50V/C0G
D26 BAV199
D27 BAV199
U15A MCP6N11-001
G=1.25/1.2=1.0417
V_TERM
R160 15K/0603
R163 360K/0603
U15B MCP6N11-001
C72 100N/0603/50V/X7R
+3V3_A
GND
U16B MCP6N11-001
C73 100N/0603/50V/X7R
R168 1K/0603
R169 1K/0603
C74 10N/0603/50V/X7R
G=7.2222
G total = 8.6667
-> 43.33mV/A
fg = 415kHz
I_OUT_P
I_OUT_N
5mV/A
D28 BAV199
D29 BAV199
U16A MCP6N11-001
R172 20K/0603
R173 100K/0603
fg = 417kHz
G=1.2
-> 6mV/A
R170 130K/0603
R171 18K/0603
U17A TLV314IDBV
R174 18K/0603
R175 130K/0603
U17B TLV314IDBV
C75 100N/0603/50V/X7R
C76 100N/0603/50V/X7R
C77 100N/0603/50V/X7R
I_OUT
9, 10, 13
R176 1K/0603
R177 1K/0603
C79 10N/0603/50V/X7R
C78 100N/0603/50V/X7R
Recalculate dividers and resulting gains!
U19B MCP6N11-001
G=2
D30 BAV199
U19A MCP6N11-001
SENS_P
SENS_N
Z=10MΩ
G=0.1
R178 750K/0805/HV
R179 750K/0805/HV
R180 750K/0805/HV
R181 750K/0805/HV
R182 750K/0805/HV
R183 750K/0805/HV
R184 1M/0603
R185 1M/0603
R187 0R/0603
R188 1M/0603
R189 1M/0603
R191 750K/0805/HV
R192 750K/0805/HV
R193 750K/0805/HV
R194 750K/0805/HV
R195 750K/0805/HV
R196 750K/0805/HV
C81 100P/0603/50V/C0G
D31 BAV199
R186 100K/0603
C80 10N/0603/50V/X7R
R190 100K/0603
C83 10N/0603/50V/X7R
R198 1K/0603
R199 1K/0603
R197 100K/0603
R200 100K/0603
U20A TLV314IDBV
U20B TLV314IDBV
C84 100N/0603/50V/X7R
I_ISO
8, 13
G=1
-> 100V/A
R201 100K/0603
R202 100K/0603
Select suitable gain
D32 BAV199
D33 BAV199
I_ISO_P
I_ISO_N
100V/A
U18 ADS131M04 TSSOP
AIN0P
AIN0N
AIN1P
AIN1N
AIN2P
AIN2N
AIN3P
AIN3N
AVDD
DVDD
AGND
DGND
CLKIN
SYNC/RESET
DRDY
SCLK
DOUT
DIN
CS
CAP
ADC
SPI
ADC 13
SPI 13
C82 220N/0603/25V/X7R
Measurement
PS1-1BA
Battery Source
Date: 2026-Mar-05
Sheet 11 of 20
Size: A3
Schema: ahorat / daniw
Layout: daniw
Measurement.SchDoc

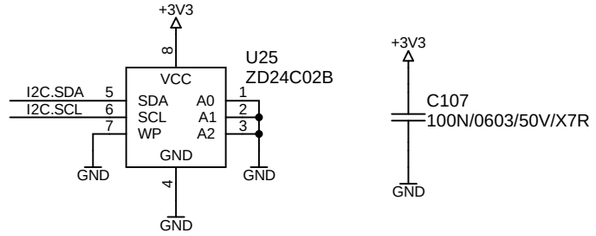

+3V3
R203
1M/0603
R204
1M/0603
Q34
BSS84
Q35
BSS84
J6
Socket_2x12
UI_PRESENT
13
+3V3_STBY
D34
B5819W SL
R205
1M/0603
GND
+15V
+15V_UI
R206
0R/0603
Q36
BSS138
BMS_ON_OFF
CHG
DSG
BMS_ON_OFF
2, 3
I2C
SCL
ALERT
SDA
2, 13, 14
I2C
R207
200R/0603
I2C_SCL_UI
R209
200R/0603
2C_SDA_UI
D35
BZT52C3V3T
D36
BZT52C3V3T
UART_TX_UI
UART_RX_UI
HV_DET_UI
+3V3_STBY_UI
R208
200R/0603
DISPLAY_MISO
R210
200R/0603
DISPLAY_MOSI
R211
200R/0603
DISPLAY_SCK
R212
200R/0603
DISPLAY_CS
R213
200R/0603
DISPLAY_WR
R215
200R/0603
DISPLAY
MISO
MOSI
SCK
CS
WR
DISPLAY
13
UART
TX
RX
13, 14
UART
R214
200R/0603
UART_TX_UI
R216
200R/0603
UART_RX_UI
D42
BZT52C3V3T
D43
BZT52C3V3T
D37
BZT52C3V3T
D38
BZT52C3V3T
D39
BZT52C3V3T
D40
BZT52C3V3T
D41
BZT52C3V3T
+3V3
R217
10K/0603
R218
100R/0603
ENCODER_B_UI
Encoder
B
A
ENCODER
13
C85
100P/0603/50V/C0G
D44
BZT52C3V3T
C86
100P/0603/50V/C0G
D45
BZT52C3V3T
9
HV_DET
R219
200R/0603
HV_DET_UI
+3V3
R220
10K/0603
R221
100R/0603
ENCODER_A_UI
R222
10K/0603
R223
10K/0603
R224
10K/0603
BUTTON
OK
ESC
OUT
BUTTON
13
Q37
BSS138
Q38
BSS138
Q39
BSS138
BTN_OK
R225
10K/0603
Ok
BTN_ESC
R226
10K/0603
Esc
BTN_OUT
R227
10K/0603
Out
BTN_OK_f
BTN_ESC_f
BTN_OUT_f
R228
0R/0603
ON_OFF
OFF
ON
UI_ON_OFF
3
D46
B5819W SL
D47
B5819W SL
D48
B5819W SL
R229
100K/0603
R230
100K/0603
D49
BZT52C3V3T
R231
100K/0603
D50
BZT52C3V3T
R232
100K/0603
D51
BZT52C3V3T
C87
100P/0603/50V/C0G
C88
100P/0603/50V/C0G
C89
100P/0603/50V/C0G
User Interface
PS1-1BA
Battery Source
Date: 2026-Mar-05
Sheet 12 of 20
Size: A3
Schema: ahorat / daniw
Layout: daniw
User_Interface.SchDoc


9, 10, 11 I_OUT
I_OUT
3 V_IN
9 V_OUT
V_IN
V_OUT
Encoder
12 ENCODER
A
B
ENCODER.A
ENCODER.B
BUTTON
12 BUTTON
OK
ESC
OUT
BUTTON.OK
BUTTON.ESC
BUTTON.OUT
BMS_CTRL
2 BMS_CTRL
I_BAT
WAKEUP
BMS_CTRL.I_BAT
BMS_CTRL.WAKEUP
11 V_TERM
V_TERM
8, 11 I_ISO
I_ISO
5 V_HV
V_HV
4, 7 OVP/
5, 8 OCP/
12 UI_PRESENT
OVP/
OCP/
UI_PRESENT
9, 10 TEMP_CURRENT
TEMP_CURRENT
14 RESET/
14 BOOT0
RESET/
BOOT0
15 V_VCC
16 V_15V
18 V_5V
17 V_3V3
19 V_3V3_A
V_VCC
V_15V
V_5V
V_3V3
V_3V3_A
U21A
PA0 20 I_OUT
PA1 21 V_TERM
PA2 22 QSPI.CS/
PA3 25 QSPI.CLK
PA4 26 VREF/2_UC
PA5 27 I_1A_REF
PA6 28 QSPI.IO3
PA7 29 QSPI.IO2
PA8 69 BMS_CTRL.I_BAT
PA9 70 I_ISO
PA10 71 CONV_CTRL.PRIM_L
PA11 72 CONV_CTRL.PRIM_H
PA12 73 ADC.CLK
PA13 76 SWD.SWDAT
PA14 77 SWD.SWCLK
PA15 78 SPI.CS
STM32G474VET
< PA4 and PA5
< available for DAC
<- Timer
U21B
PB0 32 QSPI.IO1
PB1 33 QSPI.IO0
PB2 34 HV_CTRL.EN
PB3 90 SWD.SWO
PB4 91
PB5 92 UI_PRESENT
PB6 93 ENCODER.B
PB7 94 ENCODER.A
PB8-BOOT0 95 BOOT0
PB9 96 BUTTON.OUT
PB10 47 RESTRICT: No toggle (REFBUF)
PB11 50 RESTRICT: No toggle (REFBUF)
PB12 51 HV_CTRL.PRIM
PB13 52 RESTRICT: No toggle (REFBUF)
PB14 53 RESTRICT: No toggle (REFBUF)
PB15 54 V_HV
STM32G474VET
< PB4/5 and PB6/7
< available for
< Encoder
<(TIM3 / TIM4)
R233
100K/0603
GND
U21C
PC0 15 SHUNT_EN
PC1 16 OUT_SEL_HV
PC2 17 OVP_RESET
PC3 18 OVP/
PC4 30 OCP/
PC5 31 DISCHARGE/
PC6 65 I2C.SCL
PC7 66 I2C.SDA
PC8 67 CONV_CTRL.SEC_L
PC9 68 CONV_CTRL.SEC_H
PC10 79 SPI.SCK
PC11 80 SPI.MISO
PC12 81 SPI.MOSI
PC13 7 RESTRICT: No toggle (LSE)
PC14/OSC32_IN 8
PC15/OSC32_OUT 9
STM32G474VET
Y1
Crystal 32.768 kHz HC-49/US
Y2
Crystal 32.768 kHz FC-135
C90
C91
GND
15P/0603/50V/C0G
U21D
PD0 82 ADC.SYNC/RESET
PD1 83
PD2 84 ADC.DRDY
PD3 85
PD4 86 BUTTON.ESC
PD5 87 UART.TX
PD6 88 UART.RX
PD7 89
PD8 55 TEMP_CURRENT
PD9 56 CONV_CTRL.TEMP_PRIM
PD10 57 CONV_CTRL.TEMP_SEC
PD11 58 V_3V3_A
PD12 59 CONV_CTRL.TEMP_TRAFO
PD13 60 SHUNT_ISO_EN
PD14 61 RESTRICT: No toggle (REFBUF)
PD15 62 I2C.ALERT
STM32G474VET
<==
<==
<== slow ADC
<==
<==
<==
U21H
VDD 24
VDD 49
VDD 64
VDD 75
VDD 100
VDD 37
VDDA
VREF+ 36 VREF_UC
VBAT 6 +3V3_RTC
VSSA/VREF- 35
VSS 23
VSS 48
VSS 63
VSS 74
VSS 99
VSS
STM32G474VET
+3V3
+3V3_A
+3V3_STBY
D52
BAV199
R235
1K/0603
C92 22U/0805/25V/X5R
C93 22U/0805/25V/X5R
C94 22U/0805/25V/X5R
C95 22U/0805/25V/X5R
C96 22U/0805/25V/X5R
C97 100N/0603/50V/X7R
C98 100N/0603/50V/X7R
C99 100N/0603/50V/X7R
C100 100N/0603/50V/X7R
C101 100N/0603/50V/X7R
C102 100N/0603/50V/X7R
GND
U21E
PE0 97
PE1 98 BUTTON.OK
PE2 1 DISPLAY.SCK
PE3 2 DISPLAY.WR
PE4 3 DISPLAY.CS
PE5 4 DISPLAY.MISO
PE6 5 DISPLAY.MOSI
PE7 38 CONV_CTRL.EN
PE8 39 V_3V3
PE9 40 ON_REQ
PE10 41 V_15V
PE11 42 V_VCC
PE12 43 V_5V
PE13 44 V_IN
PE14 45 OUT_SEL_ISO
PE15 46 V_OUT
STM32G474VET
<== slow ADC
<==
<== slow ADC
<==
U21F
PF0/OSC_IN 12
PF1/OSC_OUT 13
PF2 19
PF9 10 BMS_CTRL.WAKEUP
PF10 11
STM32G474VET
U21G
PG10-NRST 14 RESET/
STM32G474VET
+3V3
R234
100K/0603
CONV_CTRL
CONV_CTRL.EN
CONV_CTRL.PRIM_H
CONV_CTRL.PRIM_L
CONV_CTRL.SEC_H
CONV_CTRL.SEC_L
CONV_CTRL.TEMP_PRIM
CONV_CTRL.TEMP_SEC
CONV_CTRL.TEMP_TRAFO
EN
PRIM_H
PRIM_L
SEC_H
SEC_L
TEMP_PRIM
TEMP_SEC
TEMP_TRAFO
CONV_CTRL 4
ADC
ADC.CLK
ADC.SYNC/RESET
ADC.DRDY
CLK
SYNC/RESET
DRDY
ADC 11
SPI
SPI.SCK
SPI.MISO
SPI.MOSI
SPI.CS
SCK
MISO
MOSI
CS
SPI 11
UART
UART.TX
UART.RX
TX
RX
UART 12, 14
I2C
I2C.ALERT
I2C.SCL
I2C.SDA
ALERT
SCL
SDA
I2C 2, 12, 14
HV_CTRL
HV_CTRL.EN
HV_CTRL.PRIM
EN
PRIM
HV_CTRL 5
ON_REQ
ON_REQ 3, 15
SHUNT_ISO_EN
SHUNT_ISO_EN 9, 10
I_1A_REF
I_1A_REF 9, 10
SHUNT_EN
SHUNT_EN 9, 10
DISCHARGE/
DISCHARGE 4
OUT_SEL_HV
OUT_SEL_HV 9
OUT_SEL_ISO
OUT_SEL_ISO 9
OVP_RESET
OVP_RESET 7
DISPLAY
DISPLAY.CS
DISPLAY.MISO
DISPLAY.MOSI
DISPLAY.SCK
DISPLAY.WR
CS
MISO
MOSI
SCK
WR
DISPLAY 12
VREF_UC
VREF/2_UC
VREF_UC 14
VREF/2_UC 14
QSPI
QSPI.CLK
QSPI.IO0
QSPI.IO1
QSPI.IO2
QSPI.IO3
QSPI.CS/
CLK
IO0
IO1
IO2
IO3
CS/
QSPI 14
SWD
SWD.SWDAT
SWD.SWCLK
SWD.SWO
SWDAT
SWCLK
SWO
SWD 14
Microcontroller
PS1-1BA
Battery Source
Date: 2026-Mar-05
Sheet 13 of 20
Size: A3
Schema: ahorat / daniw
Layout: daniw
Microcontroller.SchDoc

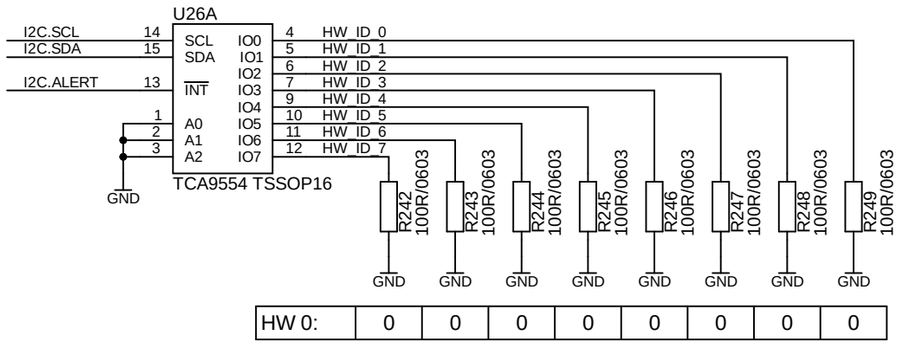


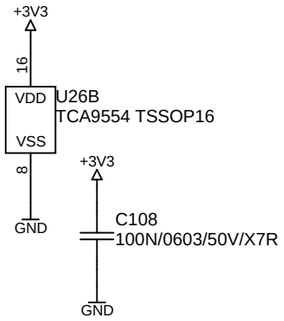

I2C Address:
0b1010000 (0x50)

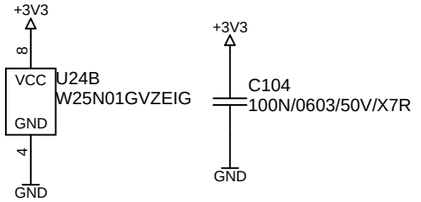


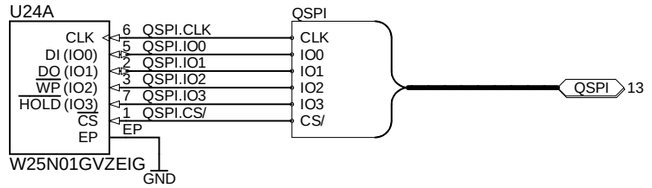

I2C Address:
0b0100000 (0x20)

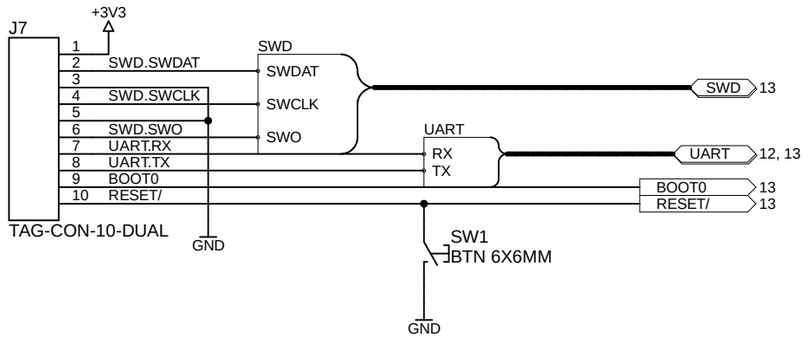


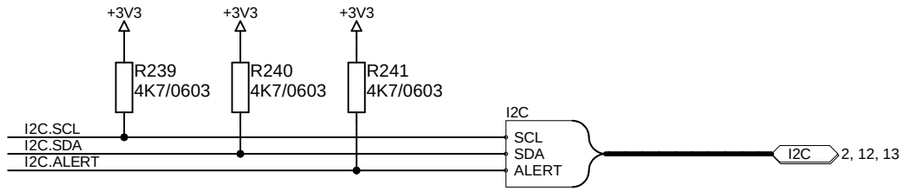

| HW 0: | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|



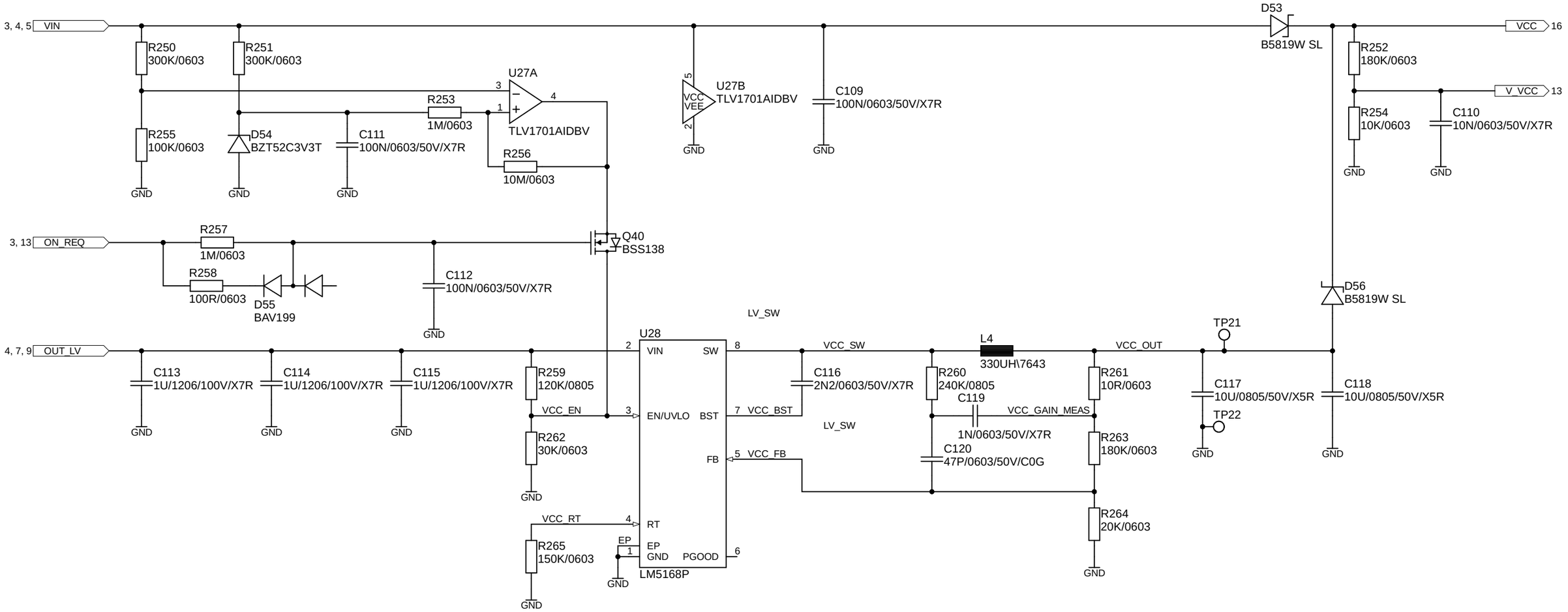



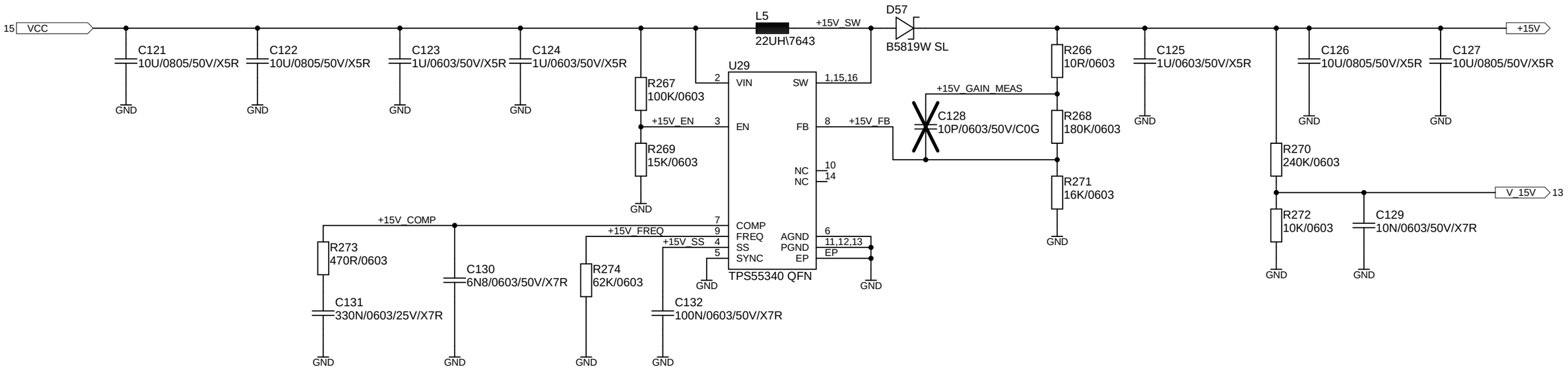



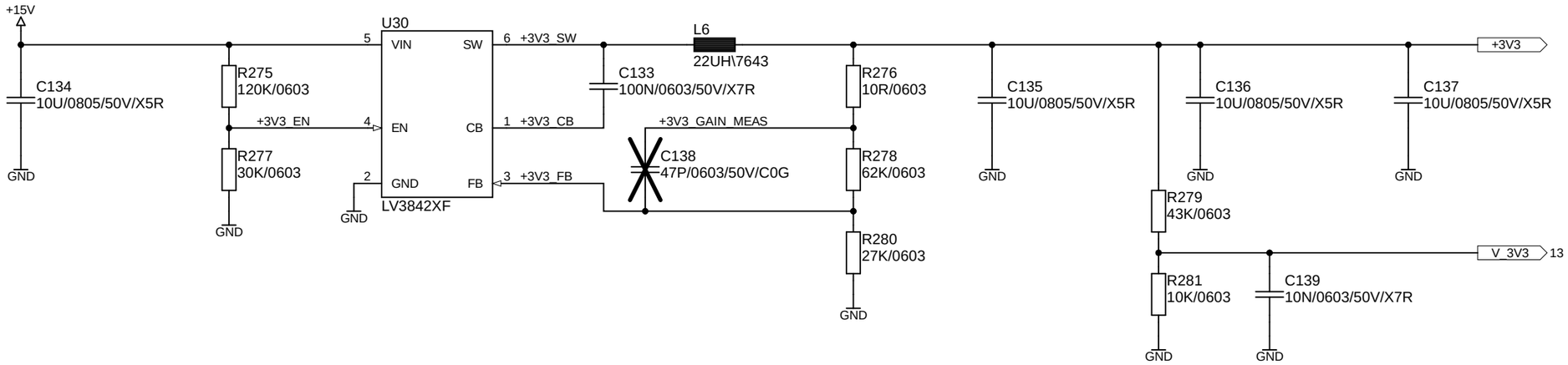

| | | | |
|---|---|---|---|
| ***Microcontroller Peripherals*** | | | * |
| PS1-1BA | Battery Source | | * |
| Date: 2026-Mar-05 | Sheet 14 of 20 | Size: A3 | * |
| Schema: ahorat / daniw | Layout: daniw | Peripherals.SchDoc | |



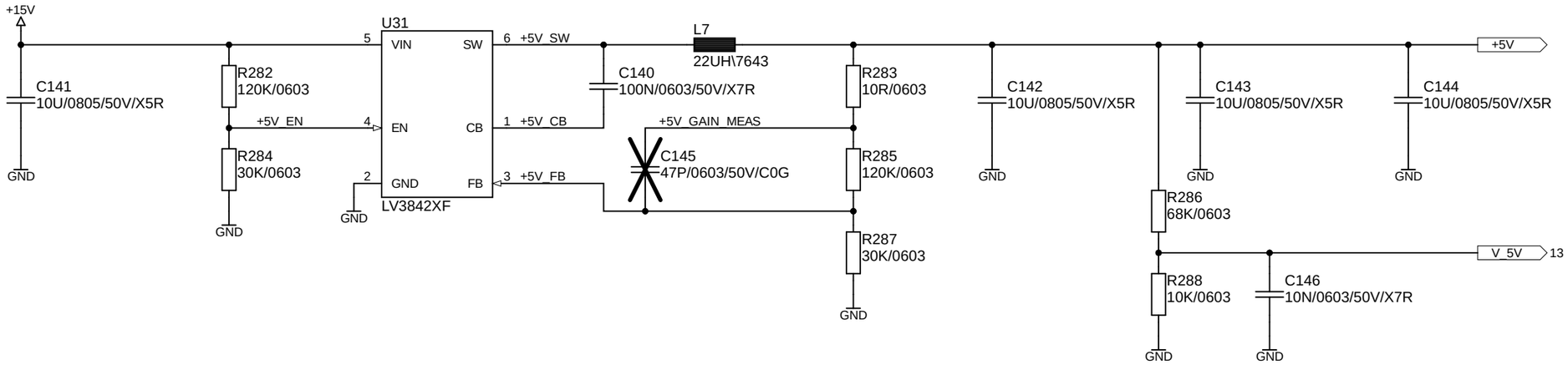

| | | |
|---|---|---|
| ***Power VCC*** | | * |
| PS1-1BA | Battery Source | * |
| Date: 2026-Mar-05 | Sheet 15 of 20 | Size: A3 * |
| Schema: ahorat / daniw | Layout: daniw | Power_VCC.SchDoc |



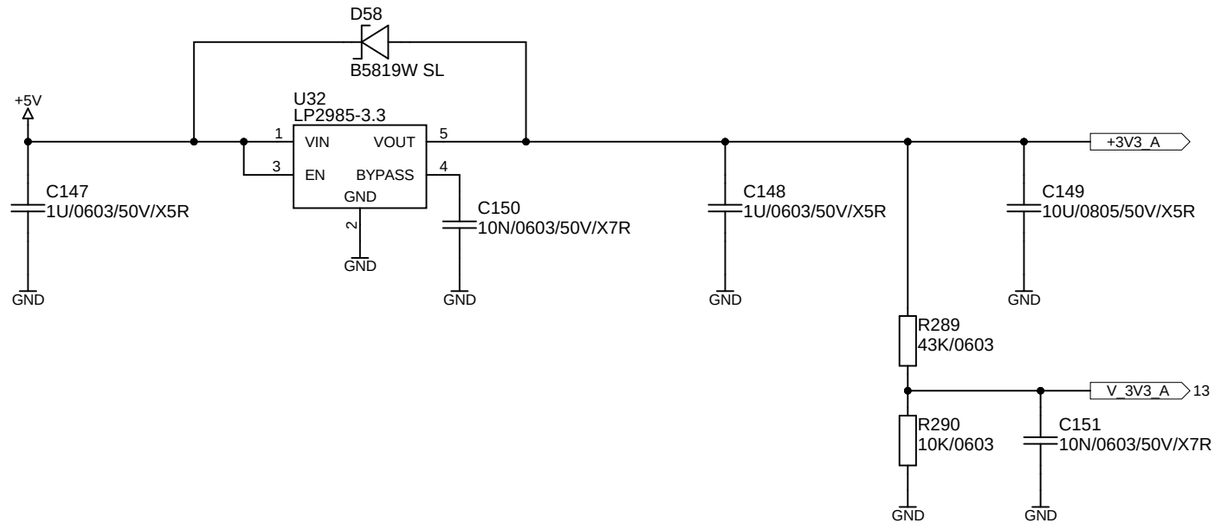

| | | | |
|---|---|---|---|
| ***Power VCC*** | | | * |
| PS1-1BA | Battery Source | | * |
| Date: 2026-Mar-05 | Sheet 16 of 20 | Size: A3 | * |
| Schema: ahorat / daniw | Layout: daniw | Power_15V.SchDoc | |




| | | | |
|---|---|---|---|
| ***Power +3V3*** | | | * * * |
| PS1-1BA | Battery Source | | |
| Date: 2026-Mar-05 | Sheet 17 of 20 | Size: A3 | |
| Schema: ahorat / daniw | Layout: daniw | Power_3V3.SchDoc | |




| | | | |
|---|---|---|---|
| ***Power +5V*** | | | * * * |
| PS1-1BA | Battery Source | | |
| Date: 2026-Mar-05 | Sheet 18 of 20 | Size: A3 | |
| Schema: ahorat / daniw | Layout: daniw | Power_5V.SchDoc | |




| | | | |
|---|---|---|---|
| ***Power +3V3 Analog*** | | | * * * |
| PS1-1BA | Battery Source | | |
| Date: 2026-Mar-05 | Sheet 19 of 20 | Size: A3 | |
| Schema: ahorat / daniw | Layout: daniw | Power_Analog.SchDoc | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

A

B

C

D

| | | | |
|---|---|---|---|
| ***Mechanics*** | | | * * * |
| PS1-1BA | Battery Source | | |
| Date: 2026-Mar-05 | Sheet 20 of 20 | Size: A3 | |
| Schema: ahorat / daniw | Layout: daniw | Mechanics.SchDoc | |

1 2 3 4 5 6 7 8